METHODS · GLOSSARY

What every column means

Every metric on the site: formulas, choices and fit statistics, updated live.

Model changelog · Submit your research

> **Editor's note:** This article was AI-generated, for transparency on methodology. A human writer has not yet edited it.

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THE CORPUS

The corpus

Everything on the site is computed from the same raw substrate. Here is what's currently loaded:

Games	23,254
Play-by-play events	6,967,972
Shifts (nonzero duration)	17,219,462
Shift summaries	18,283,586
computed_stats rows	5,597,474

DATA THROUGH **2026-10-31** · last rebuilt 2026-11-01

Data comes from the public NHL API. Regular season and playoffs are kept apart: every table opens on the regular season, and the Playoffs switch shows the postseason on its own, never pooled into the season line. Playoff shots are scored by the same xG model, which is fitted on regular-season shots only, and they feed the talent estimates. Net Rating and GSAx+ are regular-season measures and read "-" for playoff games. A nightly timer scrapes and recomputes the games played so far, so the date above is the honest extent of what is loaded -- not necessarily last night, if the timer skipped a night or a game feed lagged.

Which seasons, and why 2023-24 still draws a line

The skater and team tables, a skater's page and a team's page go back to **2010-11**. Everything else — goalies, badges, Success Above Expected, iWPA, linemates, matchups, shift performance, the forecasts — starts at **2023-24**. The database holds every season from 2010-11 and the models are fit over all of it; the line is about measurement, not storage.

The boundary is 2023-24, but not because the chips went in then. The NHL puts puck-and-player tracking at fully operational in **2021-22**, and our own shot coordinates show no break at 2023-24 to contradict that — the number of distinct coordinate values and the average shot distance both drift rather than jump.

What changes in 2023-24 is what the league *records*. Missed shots go from a steady ~27% of all attempts through 2021-22 to 30.5% in 2023-24, 33.1% the year after and 33.7% now — and the extra ones are not systematically longer or shorter, so it is a matter of more attempts being written down rather than a different kind of attempt. That changes the population a shot model is fit on, which is reason enough to split it, and it is the honest reason rather than a tidier one about telemetry. So there are two base models: the chip-era one scores 2023-24 onward, and a **pre-chip** one, trained only on earlier shots, scores 2010-

11 through 2022-23. Success Above Expected inherits the boundary directly — it is defined on `xgf_shooter_adj - xga_shooter_adj`, the headline xG pairing, and its cutoffs were calibrated on chip-era shifts, so `compute_sae.py` refuses to compute it for earlier seasons.

Sort a skater table across the line if you like; counts are counts. But a 2014-15 xG and a 2024-25 xG come from different models fit on differently recorded shots, which is why pages that show the older seasons say so.

CORSI AND ON-ICE GOALS

Corsi and on-ice goals

The classic on-ice possession proxies. While a player is on the ice at 5v5, every shot attempt the team takes is a Corsi-for (CF), every attempt the opponent takes is a Corsi-against (CA). Goals-for and goals-against (GF / GA) are the actual outcomes.

```
CF% = CF / (CF + CA)
GF% = GF / (GF + GA)
```

Implementation note: a shift's "for" / "against" attribution uses the half-open interval `(start, end]`. A shift ending exactly at the moment of an event still counts as the outgoing line on for that event; an incoming line starting at the same instant does not. This avoids double-counting at line-change boundaries, which matters most for goals: a goal stops play, so its timestamp is at once the last second of the shifts of everyone on the ice and the recorded first second of the shifts of everyone coming over the boards. Measured against an independently-built on-ice matrix, league goals-for and goals-against each match truth to within 0.1%.

Strength state is parsed from the NHL situation code (4-digit string: away goalie, away skaters, home skaters, home goalie). Empty-net states are excluded from 5v5 and the special- teams buckets.

Even strength (ES) means equal skaters with both goalies in net: 5v5, 4v4, 3v3. Time with a goalie pulled (6v5, 5v6) is not ES here, though Natural Stat Trick and the NHL's official scoring count it. Through Oct. 17, 2026, McDavid's ES on-ice goals read 19–2 here and 22–3 there; the gap is his 3–1 at 6v5 and 5v6.

THE INDIVIDUAL LINE: WHAT HE DID HIMSELF

The individual line: what he did himself

The Standard tab of Skater Stats opens on the player's own line before the on-ice numbers: what he did with his own stick and body, not what happened around him. Each is a count, so the Total / Per 60 / Per 84 toggle scales it, and each follows the strength filter from his side of the ice (a hit he took on the kill counts under PK).

A1, P1	Primary assists, the last pass before a goal, and P1 = G + A1. Secondary assists are the noisiest points in the box score; P1 leaves them out.
iSF, iCF	His shots on goal (goals included) and his shot attempts: shots, misses and blocked attempts (a teammate's block included, as Corsi counts it).
ixG, ixG(sh)	The expected goals of his own unblocked attempts, raw and shooter-adjusted. The suffix names the model, as on the on-ice columns.
iBLK	Opponent shots he blocked, as the NHL box

	score counts them. Since 2023-24 the feed also logs a teammate's shot hitting him; that is a miss, not a block, and is left out.
iHF, iHA	Hits he made and hits he took. Recorded by each rink's scorer, and rinks differ, so read them within a team before across the league.
PenT, PenD, Pen+/-	Penalties taken, penalties drawn and the difference, counted as in the penalties section below: regular season, 2023-24 on.
PP TOI, EV TOI, PK TOI	His ice time on the power play, at even strength (5v5, 4v4, 3v3) and on the kill, whatever the strength filter. Hidden by default under Columns.

xG incl. blocked shots. Every xG on the site counts unblocked attempts. The switch of that name on the Standard tab, off by default, adds the attempts an opponent blocked to ixG(sh), xGF(sh) and xGA(sh), and so to xGF%(sh) and xG±(sh), each valued where the shot was likely released: the blocked-shots model, accepted but not validated, and the same value as the Extras tab's Blk xG(sh). Only the (sh) model has that value, so the raw, (sv) and (sh+sv) columns and GAx(sh) do not move. Regular season, 2023-24 on; it is greyed out for any selection reaching earlier, for playoffs, and with score-and-venue adjustment or a danger filter on.

SCORE & VENUE ADJUSTMENT (SVA)

Score & venue adjustment (SVA)

A team that trails shoots more and a team that leads sits back, and the home team takes a little more than half the attempts whatever the score. A raw shot

share therefore mixes how a team played with the states it happened to play in. The **Score & venue** toggle on the stats tables reweights every attempt, shot, goal and xG by the state it was taken in: the home team's score differential before the attempt (seven cells at 5v5, from three or more down to three or more up; trailing, tied or leading at 4v4, 3v3 and on the power play), the venue, and the strength. After weighting, the home team's share of each metric in every state is one half (on the power play, the power-play side's share is the season's pooled share, so the power play keeps its advantage). Each state's league total is preserved exactly; only the split between the two teams moves. Column headers do not change; the toggle's state and the caption under the toolbar say the table is adjusted and which date the weights run through, and a column the adjustment does not reach is marked raw.

The textbook version is Micah Blake McCurdy's (Hockey-Graphs, 2014): with f the league's home share of a metric in a state, home attempts are weighted $\frac{1}{2}/f$ and away attempts $\frac{1}{2}/(1-f)$. Evolving-Hockey (Luke and Josh Younggren) publish their on-ice metrics on that method with a weight table per metric. We depart from it at 5v5. The team that is leading is, on average, the better team, so the raw home share in a lead state mixes the score effect with talent and understates it. We fit, per season, a binomial model of each game's home share in each state on state effects plus team-season strengths, and use the state effect alone as f; the cell is then rescaled so its total is preserved. In the pre-registered tests the talent-controlled weights beat McCurdy's on predicting the other half's goal share for every metric, twenty splits of twenty. Other strengths use the textbook weights.

```
home attempt weight  =  ½ / f*        away attempt weight  =  ½ / (1
f* = logistic(state effect), from  logit P(attempt is home's) = state
then rescaled so  Σ_home w·x + Σ_away w·x  =  Σ x   in every (season,
```

Weights are fitted *per season*, the season in progress on its games to date, refit weekly: the effects drift from year to year and all but vanished in the no-crowd 2020–21 season, and the previous season's weights, tested, do not level the current one. A state with too few attempts in a season borrows the two seasons before it for its share; at even strength the cell is then rescaled to this season's total, and on the power play the weights are normalised so that the power-play side's and the shorthanded side's season totals are each preserved exactly. There is no time-of-game term: a period split was tested and added nothing we could detect for Corsi, goals or any xG variant, which is what McCurdy found in 2014. Eric Tulsky's 2012 score-adjusted weighting and his earlier "score-close" filter (keep only the close minutes) are the ancestors; score-close tested worse than raw, because it removes most of the sample along with the bias.

Goals are adjusted too. The gain is small and real (the table), and the column is there because a toggle should be all-or-nothing on a table, not because adjusted goals are a better headline than goals. **Not adjusted**, and shown greyed under the toggle: individual counts (a player's own attempts, goals, assists, points), which are not a share between two teams; the win-probability family (iWPA, shift WPA, implied purpose, xG impact), which is already conditioned on score, venue and time; the linemates' expected xGD, whose context model has a score term; and Net Rating, which is team-relative and stays on raw inputs. Goalie danger-split columns follow in a later phase.

Pre-registered tests, 5v5 regular season 2011–12 to 2025–26, 464 team-seasons, mean R^2 over twenty random half-splits of each team's games. Score-close is Tulsky's filter; the shipped weights are the talent-controlled column.

Metric	Test (R^2)	Raw	Score-close	McCurdy weights	Talent-controlled
CF	repeatability	0.699	0.654	0.759	**0.764**
	other half's GF%	0.215	0.213	0.262	**0.269**

FF	repeatability	0.668	0.594	0.716	**0.721**
	other half's GF%	0.231	0.215	0.267	**0.274**
SF	repeatability	0.611	0.513	0.654	**0.661**
	other half's GF%	0.237	0.210	0.269	**0.276**
GF	repeatability	0.219	0.128	0.222	**0.230**
	other half's GF%	0.219	0.168	0.221	**0.225**
xGF(sh)	repeatability	0.560	0.430	0.593	**0.604**
	other half's GF%	0.309	0.270	0.323	**0.328**

CF, 2026–27 5v5	−3	−2	−1	0	1	2	3
Home attempt	0.851	0.876	0.913	0.965	1.042	1.115	1.146
Away attempt	1.209	1.164	1.093	1.038	0.961	0.895	0.874

Columns are the home team's score differential before the attempt, −3 or worse to +3 or better. Weights through Oct. 18, 2026. Thin cells (fewer than 2,000 attempts this season) pool the two previous seasons.

xGF(sh), 2026–27 5v5	−3	−2	−1	0	1	2	3
Home attempt	0.891	0.901	0.924	0.975	1.019	1.039	1.061
Away attempt	1.128	1.114	1.073	1.026	0.980	0.961	0.933

Columns are the home team's score differential before the attempt, −3 or worse to +3 or better. Weights through Oct. 18, 2026. Thin cells (fewer than 2,000 attempts this season) pool the two previous seasons.

EXPECTED GOALS (XG)

Expected Goals (xG)

A logistic regression fit on every Fenwick shot in the corpus. Target is goal vs.

not-goal. Features are geometric and contextual: distance and signed angle to net — signed because a shot from behind the goal line is not the mirror image of one from the slot — their interaction, a 26-column tensor-spline surface over distance and angle (danger is a surface over the two, not a line in each), one-hot shot type, skater-pair strength state with its own geometry deviations, and prior-event context (93 features total: 67 base features plus the 26-column spline block). The fit gives a base logit per shot; on top of it sits E, the league scoring environment for that season, and then the four xG flavors layer player identity on via Extended Kalman Filter talent posteriors. E is not optional garnish — it is the anchor that keeps the shooter- and goalie-adjusted variants honest, and it is present in all four. "Unadjusted" means no talent adjustment, not no environment: raw xG is shot quality for an average shooter against an average goalie in that season's environment, which is why raw alone runs a few percent under the power play — power-play shooters are not average, and only the shooter-adjusted variant knows that.

```
xg_unadjusted   = sigmoid(base_logit + E)
xg_shooter_adj  = sigmoid(base_logit + E + μ_shooter)
xg_goalie_adj   = sigmoid(base_logit + E - μ_goalie)
xg_fully_adj    = sigmoid(base_logit + E + μ_shooter - μ_goalie)
```

Base model diagnostics: test log-loss 0.2271, Brier 0.0612, ROC AUC 0.740, ECE 0.0069. Empty-net shots — which convert at roughly 55% and are trivially easy to separate from a shot facing a goaltender — are excluded from the model entirely, so this AUC reflects only the hard part of the problem rather than being inflated by the easiest calls. For the complete walk-through — features, calibration curves, feature ablation, EKF equations, and cross-site validation — see the Expected Goals writeup on the Writeups tab.

Shooter & goalie-and-context talent (Bayesian)

Per-player latent talent is tracked over time as a posterior on the logit scale, walked forward alongside the league environment term described above. After every shot the base model was trained on (the `is_fenwick_for_xg` mask — empty-net goals, shootout attempts and penalty shots do not update anyone, since the base model never saw them either) we update the shooter, the goalie, and Etogether. Block events are excluded from talent updates (a block never reaches the goalie). Between updates, variance inflates by a small per-shot drift to model real-world talent change. We call the goalie side of this goalie-and-context talent, not goalie talent: it is estimated from the shots a goalie faces, so it also picks up the team in front of him. About half of it is shared by both of a team's goalies in a season, and in our out-of-sample tests it added no measurable predictive value.

```
Prior:   β₀ ~ Normal(μ₀, σ²₀)
Walk:    β_t ~ Normal(β_{t-1}, σ²_walk)
Update:  p = sigmoid(base_logit + E + μ_shooter - μ_goalie)
         w = p · (1 - p)
         σ²_new = 1 / (1/σ² + w)
         μ_new = μ + σ²_new · (y - p)
```

Update for the goalie uses the opposite sign so positive `goalie_talent` means a better save-rate. Once per season boundary, shooter and goalie-and-context talent are re-centred — chance-weighted, in the odds domain, over the trailing year of shots — so the league's active shooters and goalies each net to zero. Whatever that centring removes is not discarded; it moves into E, so every

already-computed probability is unchanged at the instant of the re-frame. Within a season the frame is fixed, so a player's in-season trajectory reflects only their own shots.

Hyperparameters

μ_0 (shooter)	`-0.05 (a little below the chance-weighted average of active shooters)`
σ^2_0 (shooter)	`0.02 (SD ≈ 0.14 logits)`
σ^2_walk (shooter)	`1e-5 per shot taken`
μ_0 (goalie)	`-0.05 (replacement level, relative to the chance-weighted active league)`
σ^2_0 (goalie)	`0.0025 (SD ≈ 0.05 logits)`
σ^2_walk (goalie)	`3e-7 per shot faced (treats goalie-and-context talent as near-static across seasons)`
σ^2_max	`0.5 (variance cap)`

The goalie prior anchors to the environment term E rather than an arbitrary zero, so $\mu_0 = -0.05$ describes replacement level relative to the chance-weighted active league, not an absolute save rate. Prior mean, prior variance and walk are swept together, chosen on the pre-chip-tracking era and confirmed on the chip-tracking era, and tuned to maximize next-season predictive calibration — how well a season-end posterior forecasts next season's realized conversion rate — rather than only year-over-year rank stability. That is why both variances sit tight and both walks are small: a Bayesian posterior should be smoother than the raw rate it is drawn from, not noisier.

At each offseason boundary the raw posterior mean also shifts by a measured age-drift curve — fit from within-player season-over-season conversion deltas in

the pre-chip era and checked against the chip era — applied at full strength for goalies and at half strength for shooters, since the filter already tracks a shooter's decline from their own shots and the full curve would double-correct. The shift lands before the offseason variance bump, so it moves a posterior that is still as tight as it was in April, not one already widened for the summer.

XGF & XGA: THE SHOOTER-ADJUSTED FLAGSHIP

xGF & xGA: the shooter-adjusted flagship

Our headline shot-value stat, and — after three seasons of trying to improve on it — the plainest thing in the lineup. It is not a weighted blend of anything. It is expected goals, scored two different ways depending on which side of the ice you're looking at:

```
xGF = shooter-adjusted xG   (credit for your own shooters)
xGA = shooter-adjusted xG   (debit for their shooters)
xG% = xGF / (xGF + xGA) · 100
```

Those are variant selections, not coefficients — xGF and xGA each pick a different one of the underlying xG model's outputs, rather than summing several of them together. Both sides are shooter-adjusted, symmetric. The alternative — leaving the defensive side unadjusted, on the theory that adjusting xGA for opponents' shooter talent mostly imports strength of schedule rather than measuring defence — does not hold up empirically: shooter-adjusting the defensive side performs about the same at all situations and 5v5, and meaningfully better on the penalty kill. It also has a real hockey story behind it — a good defence works to deny dangerous shooters good looks, so someone worse ends up shooting, and shooter-adjusted xGA is the version that credits

that suppression. All four variants also carry the season's scoring environment on top of the base shot-quality model, so a season with more (or less) offense league-wide is reflected the same way in every variant — the (sh) / (sv) / (sh+sv) suffixes describe talent adjustments only. Full numbers are in the writeup.

The choice of variant comes down to one question: whose personnel does this adjustment encode, and do they persist? Shooter adjustment on xGF encodes your own shooters; on xGA it encodes theirs. Goalie adjustment on xGA encodes your own goalie; on xGF it encodes theirs. For a predictive use, keep personnel that persists and drop personnel that turns over — goalies stick with a team, shooters and opponents don't. All four xG variants (shooter-adjusted, goalie-adjusted, fully-adjusted, and plain) stay on the site as opt-in columns; which one is right depends on what you're asking. Never judge a shooter against a baseline that contains his own finishing, and never judge a goalie against one that contains that goalie — both are circular.

One caveat worth keeping in view: a skater's own xGF is not purely a shot-generation number. Shooter adjustment is applied at the shot level using whoever actually took it, so a top line whose finisher scores on low-quality chances lifts the whole line's on-ice xGF — including the playmakers who never touched the puck after the entry. Some of what looks like shot creation is a linemate's finishing talent riding along.

The name is worth separating from the metric. Tom Tango proposed weighted shots in 2014 as goals plus a flat one-fifth credit for every other attempt — no shot quality at all. What ships under xGF and xGA here is not that: it is plain expected goals, unweighted, which outperforms every weighted-shots-style blend tested against it, including talent-adjusted versions built on top of this same xG model. For the full accounting, see the writeup on the Writeups tab.

NET RATING (OFF_RTG / DEF_RTG / NET_RTG)

Net Rating (Off_RTG / Def_RTG / Net_RTG)

A per-game *Goals Above Average* rating built directly from Dom Luszczyszyn's published 2019 formula plus his recent confirmation of the offense / defense scaling factors. For the full walk-through — formula derivation, position weights, worked example, and live decomposition charts — see the Net Rating writeup on the Writeups tab.

Goalies skip all of this. Their Net Rating is GSAx — goals saved above expected on the shooter-adjusted xG model, all strengths — summed per game and projected to 84 games, which is how Dom rates them too. It is on the same goals scale as the skater number but built from a different quantity, so a goalie's +20 and a centre's +20 are not the same claim.

Step 1 — split Game Score

Standard Luszczyszyn Game Score components are split into offense and defense. Faceoff differential (0.01 weight) is left out. We record every faceoff with both participants and publish faceoff percentage, but the term needs each player's wins minus losses per game, and we do not compute that. At a 0.01 weight the omission is small. A flat 0.01 per draw treats an offensive-zone and a defensive-zone draw alike, which matters in principle. Faceoffs are not ignored on the site: the Quick Draw badge rates a centre's draws against the opponents he faced, weighted by what each draw was worth, and that rating still correlates 0.93 with raw faceoff percentage, reordering almost nobody.

```
individual_off_gs = 0.75·G + 0.7·A1 + 0.55·A2 + 0.075·SOG + 0.15·iPer
individual_def_gs = 0.05·iBLK - 0.15·iPenTaken
```

Step 2 — on-ice play-driving (5v5)

The 2019 overhaul replaced the 0.05·CF / 0.15·GF on-ice differentials with much heavier xGF / GF / xGA / GA terms, ramped per position. We measure each as the player's actual count this game minus what we'd expect for that TOI under a 50/50 mix of league and team rates per 60 (*relative-to-average* + *relative-to-team*):

```
expected_xgf  = (0.5·league_xgf60 + 0.5·team_xgf60) · TOI/3600
rel_xgf       = on-ice xGF this game - expected_xgf       (and same f
```

One asymmetry: `expected_xga` and `expected_ga` — the against-side pair only — carry a further ice-time pace factor on top of the team-pace blend above, raising the expectation for players who play heavier minutes than their position average. See “QoC/QoT, retired” below for the constants and why only the against side gets it.

Position-specific weights from the 2019 article (Forwards / Defensemen):

Position	xGF	GF	xGA	GA
Forward	0.625	0.625	1.75	0.4375
Defense	1.7	0.425	2.3	0.575

```
on_ice_off_gs(F) =  0.625·rel_xgf + 0.625·rel_gf
on_ice_def_gs(F) = -1.75 ·rel_xga - 0.4375·rel_ga
on_ice_off_gs(D) =  1.7  ·rel_xgf + 0.425 ·rel_gf
on_ice_def_gs(D) = -2.3  ·rel_xga - 0.575 ·rel_ga
```

Step 3 — QoC/QoT, retired

Through 2026-08 this step was a per-game multiplier — credit nudged up when

a player's opponents were heavier-minute than their teammates, trimmed when the opposite was true. The 2026-09 calibration sweep tested it three ways and found it predictively inert to slightly harmful, plus a face-validity failure (it charged stars for having good linemates). It has been removed. Full writeup, including what replaced it — a narrower ice-time pace adjustment folded into Step 2's `expected_xga` above, not a separate multiplier here — is in “QoC/QoT, retired” in the Net Rating writeup. That test was of our multiplier only. A second, pre-registered test in 2026-09, of an offense/defense context adjustment built a different way, is described in the same section of the writeup; the rating still carries no quality-of-competition adjustment.

Step 4 — final scaling

Subtract the position baseline (mean GS per game across F or D skaters with ≥10 GP that season), divide by total-GS-per-total-goals (≈ 2.6, computed per season), and apply Dom's offense / defense scaling factors:

```
Off_RTG (per game)  = (off_gs_game - pos_off_avg) × 0.75 / (totalGS/t
Def_RTG (per game)  = (def_gs_game - pos_def_avg) × 0.78 / (totalGS/t
Net_RTG (per game)  = Off_RTG + Def_RTG
Off/Def/Net_RTG     = Σ per-game values over the games
per 84              = Σ per-game values · 84 / GP
```

Per-game ratings live on `computed_stats` (strength="all" row), so a rating over any stretch of games is the SUM of its per-game values. On the Skater Stats table Off_RTG, Def_RTG and Net_RTG are all such sums, so in Total mode Off_RTG + Def_RTG = Net_RTG; the Per 84 and Per 60 toggles scale all three alike, Per 84 by `×84/GP`, the length of a season from 2026–27 under the new CBA. Every season Net Rating on the site, on player and team pages, the cards, the lineups and the front page, is quoted in the same unit, goals per 84 games,

so for a skater's full season the table's Per 84 Net_RTG and his season Net Rating are the same number. Until 2026-10-14 the season figures were quoted per 82; from then every value on the site, past seasons included, was the per-82 figure times 84/82, which moves no ranking, and since the 2026-10-19 rebuild the model itself computes per 84 (Σ × 84/GP), so nothing is converted on the way out.

Per-season distribution

Season	Skaters	Net_RTG/84 mean	Net_RTG/84 SD	Min	Max
2010–11	891	-2.78	9.66	-61.29	27.54
2011–12	894	-2.21	8.76	-54.50	37.89
2012–13	839	-2.41	9.46	-48.38	34.47
2013–14	886	-2.58	9.46	-63.31	40.46
2014–15	882	-2.69	9.29	-75.01	40.91
2015–16	898	-2.57	9.63	-59.53	70.45
2016–17	888	-2.87	8.84	-61.95	23.02
2017–18	890	-2.48	9.78	-73.62	49.39
2018–19	906	-3.00	10.52	-83.59	98.73
2019–20	883	-2.56	10.56	-111.11	44.15
2020–21	913	-3.48	11.07	-60.74	35.07
2021–22	1,004	-3.57	11.22	-61.27	45.39
2022–23	951	-3.46	11.52	-67.64	56.84
2023–24	924	-2.81	10.93	-89.29	45.44

2024–25	920	-2.89	11.01	-58.02	58.95
2025–26	940	-2.99	11.37	-115.51	62.20
2026–27	740	-3.13	13.90	-82.21	57.68

The mean sits below zero in every season, and that is an artifact of how the table is built rather than a finding about hockey: the population is *unfiltered* — every skater who recorded a shift is in it, including several hundred a year on call-up minutes, and marginal players sit well below average by construction. Across sixteen seasons it stays between −2.2 and −3.6, wandering rather than trending, and there is no step at 2023-24 where the puck-tracking data begins.

The spread is the part worth reading, and it is stable: the SD runs 9–12 across sixteen seasons with no trend, and the tails are intact throughout. Restricted to skaters with at least 20 games the means are far closer to zero — −1.01, −0.72 and −0.82 for 2023–24, 2024–25 and 2025–26. We cannot show a minutes-based column here at all: `player_ratings.toi_5v5_seconds` is unpopulated in every row of every season and nothing writes it, so the restriction above is on games played.

Those figures are for the sixteen complete seasons. While a season is still being played its row comes last and runs wider, because every rating in it is a per-84 rate on a handful of games. Cut 2025–26 at each skater's first few games and the SD is 18.2 at three games, 15.7 at five, 13.6 at ten, 12.3 at twenty and 11.5 at forty-one, against 11.4 for the full season: the row narrows into the range above as the games accumulate.

Caveats

- **Defensemen run slightly over-corrected** by the ice-time pace adjustment on `expected_xga` — Def now correlates +0.13 with TOI for defensemen, versus 0.00 for forwards. Within noise; being watched, not yet re-fit.
- **Team pace** (a team's overall xG production) is handled by the team-relative half of the `rel_*` calculation in Step 2. **Individual ice-time pace** (a player's own minutes) is the separate against-side factor described in Step 3. They are two different adjustments; don't conflate them.

In four parts

The Standard tab of Skater Stats can break Net Rating into four columns that add up to it, under **Net Rating parts** in the column picker. **Individual** is his own Game Score terms: goals, assists, shots, penalties drawn and taken, blocks. **On-ice for** and **On-ice against** are the 5v5 xGF(sh) and goals for, and xGA(sh) and goals against, each against expectation. **Context** is the ice-time pace credit plus the penalty-kill context term. Each part is measured against the average skater at his position for that part, so the position baseline is split among them rather than standing apart. They are season sums like Off_RTG and Def_RTG: at Totals they add to Net_RTG, at Per 84 to his season Net Rating. Whole regular seasons at All situations only. The worked example in the Net Rating writeup lists every line behind them.

ESTIMATED NET RATING

Estimated Net Rating

A forecast of a skater's Net Rating, made before the first game and moved by every game after it. It starts from the last three seasons (each aged to this one and weighted by its games), pulled toward the average of players with that much history, and it hands over to the season's own games at a rate set by how much talent actually moves between seasons. Two numbers come out of it, kept apart on purpose: the *talent* (the per-game rate we think he produces at, times 84) and the *year-end projection* (the games already played plus the talent for the rest). The cards and the lineup print the year-end number with its 80% band, and the band is the predictive one: it allows for the noise of the games still to come, and it was checked against realised seasons. For the derivation, the holdout against the obvious baselines, the level check and its known misses, see the Estimated Net Rating writeup.

Goalies are not forecast here; the lineup card gives them their next-season forecast GSAx, scaled to 84 games at the league's measured shots faced per game.

SHOOTING METRICS

Shooting metrics

The Snipers report puts a shooter's goals next to their individual expected goals on each model and takes the difference:

iXG	Individual expected goals: the sum of xG over the player's own shots. The suffix names the model, same as on-ice xG. None for raw shot quality, (sh) shooter talent, (sv) goalie-and-context talent, (sh+sv) both.
GAx	Goals minus iXG on the raw model. Finishing above what shot quality alone predicts; a good shooter should run positive here for a career.
GAx(sh)	Goals minus iXG(sh). Finishing above what the shooter's OWN talent posterior and shot quality predict. This is the regression column: a large positive is a hot streak the model does not believe in, a large negative is a slump on a shooter it still rates.
GAx(sv), GAx(sh+sv)	The same difference against the goalie-adjusted and fully adjusted models, for readers who want the goalies faced accounted for.

GAx is not GSAx with the sign flipped: GSAx uses the on-ice xGA a goalie faced, GAx uses only the shots the skater took. The shooter's talent posterior itself, the

thing GAx(sh) is measured against, is not a column on the Skaters table. It has its own dashboard, Shooter/saver talent, with its uncertainty and the year of shots behind it.

PDO AND LINEMATE LUCK

PDO and linemate luck

PDO is the oldest luck stat in hockey: on-ice shooting percentage plus on-ice save percentage. The league's adds to 100 by construction, and a skater far from it is usually riding, or fighting, the bounces. Its trouble is that it lumps three people together: the player, the teammates shooting beside him and the goalie behind him. The Luck tab pulls him out of it, then asks what his points would have been had his teammates finished like everyone else.

oiSH%, oiSV%, PDO	100 × GF ÷ SF and 100 × (1 - GA ÷ SA) while he was on the ice; PDO is the sum. The traditional numbers, him included.
Tm SH%	100 × (GF - his goals) ÷ (SF - his shots). His teammates' shooting percentage with him on the ice, his own shots taken out.
Tm xGF	On-ice xGF minus his own iXG, on the raw model: what league-average finishers would have scored on his teammates' shots. No suffix, because nothing has been adjusted for: these are "normal" linemates.
Tm GAx, Tm dFSh%	Teammates' goals minus Tm xGF, as a count and per 100 of their unblocked attempts.
Tm xGF(sh), Tm GAx(sh)	The same on shooter-talent-adjusted xG: his teammates at their own finishing talent. A big

	Tm GAx with a small Tm GAx(sh) means his linemates can really shoot; big on both means they ran hot.
Own GAx	His goals minus his own iXG on the raw model: his own finishing, the other source of luck, shown apart.
xA1, xA2, xA	His primary and secondary assists had his teammates scored Tm xGF instead of Tm GF, at his own rate of assists per teammate goal.
xPts, xPts − Pts	His points with his own goals kept and his assists replaced by xA. Above zero, his linemates' finishing cost him points.
xPts(sh), xPts(sh) − Pts	The same with his teammates at their own talent: the luck out, their real shooting left in.

```
T_G  = GF - iG                          teammates' goals
T_X  = c × (xGF - ixG)                  at league finishing, c = league
a1   = (A1 + k1 × m1) ÷ (T_G + k1)      his primary assists per teammate
a2   = (A2 + k2 × m2) ÷ (T_G + k2)      the same for secondary assists
xA   = A1 + A2 + (a1 + a2) × (T_X - T_G)
xPts = Pts - A + xA
```

With no shrinkage, xA is simply his assists scaled by his teammates' expected goals over their real ones. The shrinkage is there for the small seasons: m is his position's rate (forwards or defencemen) in the same season and situation, and k is how many teammate goals of that average stand behind his own, measured from the spread of real assist rates. A season with no teammate goals reads at the position's rate, not 0 ÷ 0. The scale c is the league's own finishing in that season and situation, so league-average finishing reads zero in a year when the

whole league shot cold.

At All the tab counts even strength, the power play and the penalty kill, and leaves empty-net play out, because an empty-net shot carries no xG. Empty-net points he did score are kept in Pts and xPts as they are.

The prior. Assists per teammate goal are mostly noise from one season to the next. Measured across past seasons, a 5v5 forward's primary-assist rate behaves as if 86 teammate goals of his position's average stood behind every season, and his secondary-assist rate as if 428 did. So a season with a couple of dozen teammate goals is read mostly at the position's rate, and a season with none reads exactly that.

Does it add up? Across the league, expected assists should match real ones. In 80 season-and-situation cells, 2010–11 to 2025–26, 0 missed by more than the larger of 2% and two standard errors (chance alone would allow up to 7); the largest miss was 0.77 standard errors. Pooled over every season, 5v5 missed by +0.10%; All situations missed by −0.12%. Without scaling expected goals to each season's league finishing, 7 cells would have failed.

Does it predict? Skaters with 40 games in consecutive seasons: how well does this season's scoring rate call next season's?

Points per 60	Players	r, Pts	r, xPts	r, xPts(sh)	Gain in r [95%]	Gain, team changers
5v5	6,745	0.718	0.725	0.730	+0.006 [+0.002, +0.012]	+0.012 [-0.002, +0.024]
All situations	6,745	0.815	0.820	0.824	+0.004 [+0.001, +0.008]	+0.010 [+0.000, +0.019]

Linemate-normalised points call next season a little better than raw points, at 5v5 and in all situations, and out of sample too. A little: most of a scorer's scoring is his own.

Is it luck? If teammates' finishing is luck, it should not follow a player who changes teams. It does not: year to year, r = +0.035 [−0.016, +0.087] for the 1,444 who moved, against +0.181 for those who stayed with the same teammates. His own finishing follows him more, r = +0.101 after a move: some of that is skill, which is why Own GAx is a column of its own and not folded into xPts.

Pre-registered before any of it was run (commit 5c64a827): the method, the gates and the samples.

What is not tested: that he would have assisted on his teammates' missing goals at his usual rate. That is the method's assumption, and the reason xPts is an estimate, not a fact.

FO VALUE: WHAT A FACEOFF WIN IS WORTH

FO value: what a faceoff win is worth

FO% says how often a centre wins the draw. FO value says what those wins were worth, in goals. A won draw is worth more in some places than others: an end-zone draw at five on five, a power-play draw in the attacking zone, a draw with the goalie pulled.

What a win is worth. For every regular-season faceoff we count goals for minus goals against, from the side that won, in the 20 seconds after the drop, cut short by the next faceoff or the end of the period. The value of a win in each zone and strength is the average for the side that won minus the average for the side that lost, over five regular seasons (2021–22 to 2025–26, 370,890 draws). At five on

five an end-zone win is worth about one goal per 63 wins, and a neutral-zone win about one per 193. A power-play win in the attacking zone is about one per 33, and a win with an extra attacker about one per 19. An offensive-zone win and a defensive-zone win are the same draw seen from opposite sides, so they share one value. Rare states (four on three, six on four) count as draws but are worth nothing. The table is fitted once and changes only when we refit it on purpose.

A centre's FO value. His win rate is shrunk toward 50%: wins plus *a*, over draws plus 2*a*, where *a* is fitted each season from how widely the takers with 300 or more draws are spread. In the first weeks of a season, before 100 takers reach 300 draws, the five seasons' prior is used instead. FO value is that rate minus 50%, times the value of each draw he took, summed over the season.

```
rate = (wins + a) / (draws + 2a)
FO value = (rate - 0.5) × Σ value of each draw he took
```

It is a count in goals, like FOW. Total is the season, and Per 84 scales it by 84 over games played, like every count. At a strength filter it counts only the draws he took at that strength, read from his side the way FOW and FOL are, with the season's one win rate. It is computed after each night's games are final. Regular season, whole seasons only: it is blank on playoff views, with a date window, and for anyone who took no draws.

It does not adjust for the man across the dot. It also gives the taker the whole draw, as the play-by-play does, including when a winger won the puck after the drop. It measures what the draws were worth, not how good the centre is.

Method: JP Gambatese, 'What is a faceoff win worth?' (NJ Devils Advocates, 2026), reimplemented on Post Hockey's play-by-play. On the same five seasons our value table matches every goal value he published to within 0.0001 of a goal.

This stat came from outside research. Have yours? Send it to /contribute.

Penalties drawn and taken, against expected

Some skaters hand their team several more power plays a season than others, and some give several more away. The Penalties columns on the Skater Stats Extras tab count them and set them against what a league-average skater would have drawn and taken in the same minutes.

What counts. PenD is penalties drawn, PenT penalties taken: minors (a double minor once), majors other than fighting, match penalties and penalty shots. Fights and misconducts are left out: they seldom put either team on the power play. A penalty counts whether or not he was on the ice when it was called. A delayed call counts at the strength the game was actually at, not at the six on five of the pulled goalie.

Expected. xPenD and xPenT are his minutes times the same season's league rate for a skater of his position, forward or defenceman, in the same circumstances: the strength, home or away, the score (leading, tied or trailing by one or by two or more) and the period. All three move the rate. At five on five a skater leading by two or more takes about 40% more penalties than one trailing by one, the home side takes fewer and draws more, and the second period has the most calls. The rate is the season's own: the league's rate moves five to ten percent from one season to the next, and last season's rate missed by that much. During a season it is the season so far, recomputed after each night's games.

```
xPenD = Σ his seconds × league drawn rate (F/D, strength, venue, scor
PenD vs exp = PenD - xPenD          PenT vs exp = xPenT - PenT
Pen+/- vs exp = PenD vs exp + PenT vs exp      (positive is good on a
```

No role term. Depth players take more penalties a minute than top-line players do (bottom-third minutes: about 13% more than expected; top third: about 6% fewer). We leave that in. It is a real difference between players, not a fault in the baseline, so PenT vs exp favours the players who play the most minutes, and is meant to.

How much is skill. At five on five a skater's penalties against expected repeat from one season to the next: the correlation is 0.58 for PenD vs exp and 0.64 for PenT vs exp. On the power play or the penalty kill alone it is 0.14 to 0.22, so the PP and PK views are shown but are mostly noise. Read them at All, ES or 5v5.

Where it applies. Every strength aggregate (All, ES, 5v5, PP, PK); a single state such as 5v4 shows "-". It follows the date window and the rate toggle like any count, and rows that pool seasons add up. Regular season, 2023-24 on.

These are not the penalty counts Net Rating and Game Score+ read, which take every penalty called while he was on the ice, fights and misconducts included. GS+ also has its own penalty expectation, from last season's league rates and the player's own record, which is a different number from xPenD and xPenT; the GS+ writeup calls it the GS+ penalty expectation.

GOALIE METRICS

Goalie metrics

Goalie columns are split across three presets — Value, Stop rate and Tiers. As with xG, the suffix says what has been adjusted for: `(sh)` shooter talent controlled, `(raw)` nothing controlled.

GSAx(sh)	xGA on the shooter-adjusted xG model - GA. Positive = stopped more than the shot-quality-and-shooter-talent model expected. The

	headline goalie number.
GSAx(raw)	xGA on the raw (unadjusted) xG model - GA. The same idea without controlling for who was shooting.
GSAx+	IQ-scaled (mean 100, SD 15) version of GSAx(sh), computed across the full league of goalies for the season (≥10 GP). 100 is league average; useful for comparing goalies across seasons of different shot-quality environments.
Save talent	Latest Bayesian posterior mean for the goalie — goalie-and-context talent, see the Shooter & goalie-and-context talent entry above — on logit scale, relative to the active league and re-centred each season. Positive = above-average save rate given shot quality.

Goalies also get a per-player profile chart (*Implied Save Talent*) analogous to the shooter version, plotting the posterior mean ±1σ ribbon over the games they've faced.

Stop rate

Fenwick save percentage instead of the traditional shots-on-goal save percentage — it counts missed shots along with shots on goal, which is the more stable denominator and the one the rest of the site's on-ice numbers already use.

Fsv%	1 - GA/FA. Actual Fenwick save rate, no model involved.
xFsv%(sh)	1 - xGA(sh)/FA. What the shooter-adjusted xG model expected the goalie's Fenwick save rate to be, given the shots and shooters faced.

dFsv%(sh)	Fsv% - xFsv%(sh). The same signal as GSAx(sh), expressed as a rate instead of a count — positive means outperforming the shot-and-shooter-quality model.
Fsv%(raw), xFsv%(raw), dFsv%(raw)	The same three, built on the raw (unadjusted) xG model instead.

Tiers

The Save talent column is a posterior mean; the walker also carries a posterior SD, and the Tiers preset draws the two together. Each row gets a small graph on one shared logit axis: a thick bar for the mean ±1σ, a hairline for ±2σ, a dot at the mean and a vertical tick at zero, the league mean. The letter is read off the ±1σ interval [lo, hi] against that tick, where “mostly” means more than three-quarters of the interval on one side:

A	Interval entirely above the league mean: lo > 0.
B	Mostly above: lo ≤ 0 but the mean sits more than half an SD above zero (mean - σ/2 > 0).
C	Straddles: the mean is within half an SD of zero (|mean| ≤ σ/2).
D	Mostly below: the mean sits more than half an SD below zero, but hi ≥ 0.
F	Interval entirely below the league mean: hi < 0.
SD	The posterior standard deviation itself, on the same logit scale as Save talent.

A letter is a statement about confidence as much as ability: a starter with a long

record and a C is genuinely average; a call-up with a C is unknown. The graph is there so the two are not confused.

Splits

Every shot a goalie faces can be sliced four ways at once, and each slice carries the same stop-rate triple as above: FA (the Fenwick shots in the cell), Fsv% (his actual rate on them), xFsv%(sh) (what the shooter-adjusted model expected) and dFsv%(sh), the difference -- positive is stopping more than the shots in that cell predicted.

By danger	HD is a raw xG of 0.15 or more, LD is below 0.06, MD is everything in between.
By side (goalie's view)	Left and Right are more than 7 ft off the centre line -- from the GOALIE's own point of view, not the shooter's -- and Middle is within 7 ft either way.
By shot type	Tip and Deflection are their own canonical shot types; Straight is everything else -- wrist, snap, slap, backhand, wrap-around, poke, bat and shots simply typed “other.”
By play	Rebound is a shot within 3s of the same team's previous unblocked shot -- the xG model's rebound flag. Other is any shot that is not a rebound.

xReb is the number of rebounds a league-average goalie would have allowed on the same saves, from a model fitted once on every complete regular season since 2010-11 (distance, angle, raw xG, shot type, strength, period, score state and danger band), with each season's level recalibrated so that RebAx sums to zero league-wide within a season; it is a comparison with that season's goalies, not across eras -- the league rebound rate recorded in the feed stepped from

about 7.5% to 12% after 2021-22. **RebAx** is rebounds allowed minus xReb -- positive means he gave up more rebounds than the chances in front of him predicted. **Reb GSAx(sh)** is GSAx(sh) restricted to rebound-chance shots only.

Contact %

Contact % asks a different question of the same shots: how often did the goalie get a piece of the puck? A *contact* is a save, or a miss that he froze within two seconds. A *failure* is a goal, or a post or crossbar that he did not freeze. A frozen post is left out, as is any shot from outside the offensive zone; the denominator, shown as Contact shots, is contacts plus failures. C% is contacts over that denominator, xC%(sh) is what the shooter-adjusted xG model expected it to be, and dC%(sh) is the difference. CAx(sh), contacts above expected, is the same difference in counts: the contact failures the shots predicted (from shooter-adjusted xG) minus the ones he allowed, so positive is good, like GSAx. It counts posts, so it is not goals saved above expected. Posts counts the unfrozen posts, xPosts is what the shots predicted from a model fitted once on complete seasons and recalibrated within each season, and Posts over xP is the difference: positive means he was beaten to the post more often than the shots predicted, which is worse. Shots from outside the offensive zone are left out, a change suggested to us before the test was run: about one shot in forty, and almost never a goal, so the in-zone ledger matched the original on repeatability and was the form whose carry-over among goalies who changed teams cleared zero once adjusted for shot quality. Tips and deflections stay in; leaving them out made it less repeatable.

The idea is Aaron Knodell's (Puck Over the Glass, August 2026), who proposed it. Our replication found that it repeats from one year to the next better than save percentage and better than dFsv%(sh). The signal is in the posts, and it follows the goalie when he changes teams. The frozen misses are scorer behaviour, a property of the rink or the team rather than the goalie, and add

nothing. One test did not come out. We pre-registered whether the whole metric's carry-over, among goalies who changed teams, could be told apart from zero, and at our 207 such pairs it could not across all situations; at 5v5, the situation the original article used, it could. Contact % is published on the overall evidence, by decision, not by the rule. Posts over xP did clear its own test among the goalies who moved. The full numbers are in `backend/research/goalie_contact_pct.md` and `backend/research/goalie_contact_pct_followup.md`.

This stat came from outside research. Have yours? Send it to /contribute.

A column marked with an asterisk (*) is not proven to be repeatable at this table's sample size: split a goalie's games into odd and even halves, and one half did not predict the other beyond chance. Hover the header for the split-half correlation, its confidence interval, and the full-season reliability behind the star.

STANDINGS & TEAM RATINGS

Standings & team ratings

The Standings page prints four tables, one at a time. The record is what happened. The deserved columns ask what should have happened, if goals and the chances behind them were all that counted. The ratings are six answers to “who’s number one?”, and they disagree on purpose: each one decides differently what a win, a margin or a schedule is worth. Colley, Keener, Markov and the offense-defense rating are from Amy Langville and Carl Meyer’s *Who’s #1? The Science of Rating and Ranking* (Princeton, 2012), and so is Massey’s least-squares method, of which the Simple Rating System, Sports-Reference’s, is the version without home ice. Elo is the sixth. The Consensus column is the plain average of the six ranks, not a seventh opinion. Early in a season Markov,

which ranks teams on a handful of results, is withheld, and Consensus is the average of the other five. The chances are the season simulation's: what is still to come.

P%	Points divided by points available: (2W + OTL) / 2GP.
L10	The last ten games: wins, losses, overtime losses.
Strk	The current streak.
Proj PTS	Projected final points: points so far plus the expected points of every remaining game from the dressed lineups' strengths (the same projection as the team cards). Hover for the 80% band.
RgPts%	Regulation points percentage: every game that went past regulation scored as a tie, one point each, the way the standings were kept before the shootout. (2RW + overtime games) / 2GP.
PythPts%	Pythagorean expectation: goals for to the k divided by the sum of goals for and goals against to the k, on goals excluding shootout goals, with k fitted on every season since 2010-11; moved onto the points scale by adding half of the league's overtime share for each expected loss.
Pyth W%	The Pythagorean win share itself, before the move onto the points scale.
xW%	Expected winning percentage: the Pythagorean expectation on expected goals, xGF(sh+sv) to the k over the sum of xGF(sh+sv) and

	xGA(sh+sv) to the k, with k fitted on every team-season since 2015-16 so that teams which out-chance their opponents at a given rate win as often as the number says. Regulation and overtime chances; no shootouts.
xPts%	xW% on the points scale, moved there the way PythPts% is: half of the league's overtime share added for each expected loss.
SRS	Simple Rating System: average goal margin plus average opponent rating, solved so every team's rating is its margin against an average schedule. Goals in regulation and overtime; a shootout is a zero margin. League average 0.
MOV	Average goal margin per game, shootouts as zero.
SOS	Strength of schedule: the average rating of the opponents faced (SRS minus MOV).
Elo	Elo rating, replayed over every regular-season game in the database, carried between seasons toward 1500, with the margin of victory weighting the update and a home-ice term in the expectation. 1500 is average.
Massey	Massey's least-squares rating (Who's #1?, ch. 2): the goal margins explained by a rating for each team plus one home-ice term. Without the home-ice term it is SRS exactly.
Massey Off / Massey Def	Massey's split of the rating into an offensive and a defensive part (rating = off + def), each in goals per game against average; higher is better for both.
Colley	Colley's rating (ch. 3): wins and losses only,

	Laplace-smoothed, each opponent's rating feeding back into yours. A game past regulation is half a win for each side. League average 0.500.
Keener	Keener's rating (ch. 4): the share of the goals in each game, smoothed and skewed so a blowout counts little more than a clear win, with the leading eigenvector of that matrix as the rating. League average 1.00.
Markov	The Markov rating (ch. 6): every team votes for the teams that scored on it, goal by goal, and the rating is where a random walk along those votes spends its time. League average 1.00. Withheld early in a season, while it ranks teams on a handful of results.
OD / OD Off / OD Def	The offense-defense rating (ch. 7): offence is the goals scored weighted by how hard each opponent is to score on, defence is the goals allowed weighted by how good each opponent is at scoring, solved together. OD = Off / Def. Off above 1 is good; Def BELOW 1 is good.
Consensus	Mean rank over Elo, SRS, Colley, Keener, Markov and OD; over the other five while Markov is withheld.

Cup chance EXPERIMENTAL

The chance a team wins each playoff round, and the Stanley Cup, as the share of the season's simulated seasons in which it happened. Each simulated season ends in a bracket in the NHL's format: three teams from each division and two wild cards per conference, best of seven with home ice 2-2-1-1-1, the conference final and the Final by record. Each team plays its playoffs at the strength it ended that simulated season with. In playoff games the regular-season strength gap

between two teams is multiplied by c = 0.479, fitted by maximum likelihood on 1,048 playoff games from 2013-14 to 2025-26, without 2019-20; left out one season at a time, c runs from 0.40 to 0.58. With the full gap, the stronger team won past series less often than the model said it would. Experimental: the 2026-27 playoffs are the out-of-sample test, and the numbers come down if April contradicts them.

SELKE RACE

Selke race

The Selke Trophy goes to the league's best defensive forward, and the case for a candidate is usually a mix of eyeball, faceoffs, and penalty-kill minutes — three things a box score reports and none of which is actually a defensive-impact number. This preset puts those alongside the one column on the site that is: **Def_RTG**, the defensive half of Net Rating.

Forwards only, sorted by Def_RTG. Faceoff win % (FO%, now filed under Extras rather than Identity) is draws won over draws taken, counted from the play-by-play at the table's strength, with the counts themselves beside it as FOW and FOL (none of it derived from Def_RTG or any xG model) — it is on this preset because voters look at it, not because we think it measures shutdown play. PK TOI is minutes played shorthanded, another usage signal rather than an impact one; it now sits right next to TOI and the new PP TOI column, since all three are the same kind of number (minutes played, sliced by strength) rather than a faceoff stat. PP TOI — minutes played on the power play — isn't part of the Selke case, but it's built the same way as PK TOI and lives in the same column group for symmetry. Def_RTG is the column that answers whether the ice actually tilted the right way while they were out there.

Quality of Competition / Teammates

There are *two* QoC/QoT measures on this site, and they answer different questions. Keep them apart.

Competition (usage) — how heavily the opposition is played

TOI-based: for each player, compute the average game-level TOI% of their on-ice teammates and opponents across all 5v5 shifts they played, weighted by shift TOI when rolling up to (player, season). This is a *deployment* proxy — it measures how much a coach trusts the players you are on the ice against, not how good they are.

```
toi_pct[player, game] = player_5v5_toi / sum(team skaters' 5v5 toi ir
shift_qoc = mean(toi_pct of opponents on ice)
shift_qot = mean(toi_pct of teammates on ice)
season    = TOI-weighted mean of shift values
```

This is the usage measure that used to be folded into Net Rating as a per-game context multiplier, from launch through 2026-08 — see “QoC/QoT, retired” in the Net Rating writeup for why that stopped in 2026-09. The measure itself is unaffected and still exported as a CSV at `backend/exports/qoc_qot_<YYYYMMDD>.csv` for offline analysis.

Competition (Net_RTG) — how good the opposition is

A *talent* proxy, and the axis the Shift Performance dashboard slides along. For each shift, average the season Net Rating of the opposing skaters on the ice; do the same over the player's own skaters for teammates. Goalies are excluded from both.

```
rating[player, season] = mu + (net_rating - mu) * gp/(gp + 20)
shift_qoc = mean(rating of opposing skaters on ice)
shift_qot = mean(rating of own skaters on ice)
```

The input is plain Net Rating. Until 2026-09 it was a second column, net_rating_unadj, holding Net Rating with the QoC/QoT multiplier divided back out — defining QoC in terms of a number that already contained a QoC adjustment would have been circular. With that multiplier retired the two were the same number, so the second column is gone. Ratings are **shrunk toward the season mean** by gp/(gp+20) before being averaged, because season Net Rating is a full-season projection: without shrinkage a four-game call-up projects to an absurd value and swings the QoC of every shift they appear in. QoC and QoT are on the same per-84 scale as the Net Rating published everywhere else; until the 2026-10-19 rebuild they were per 82 and read 2.4% smaller.

Stored per shift in `shift_qoc_facts` (its 5v5 part only, ~12.4M rows), which is what lets the Shift Performance dashboard re-aggregate a whole stat line from an arbitrary slice of shifts instead of re-fitting a model. The dashboard filters these ratings by **decile** — ten equal-count bands of the season's 5v5 shifts, 1 the softest competition (or weakest teammates), 10 the hardest — while the xG impact and WP impact forecasts cut the same ratings into **terciles**over every 5v5, power-play and penalty-kill shift; same numbers, two grains. Both the shrinkage constant and the ≥3-of-5 rated-opponent coverage rule are provisional.

IMPLIED PURPOSE

Implied Purpose

A coach does not send five players over the boards without a reason. Down one with four minutes left you get your scorers; up one with thirty seconds you get your checkers. *Implied purpose* is an attempt to read that intent off the game state — what the bench was trying to do — so that a shift can be judged against its own brief rather than against a league-average one.

It is the filter on the Shift Performance dashboard, and it is an **inference, not a record**. Nobody publishes what the coach intended. One observable thing stands in for it: how lopsided the stakes are at the moment the shift starts, and again whenever the strength changes while it is out there. A shift that runs into a power play is read twice, once at five-on-five and once on the power play.

```
wp = (dwp_for - |dwp_against|) / (dwp_for + |dwp_against|)
score = wp + 0.05 * lineup tilt (z, clipped) + 0.03 * zone start (OZ
top third of all shifts → attack;  bottom third → defend;  middle -
```

Leverage. From the win-probability model: how much a goal for would help, set against how much a goal against would hurt, given the score, the clock, the strength state and any penalty time still to run. Late and trailing, a goal for is worth far more than a goal against costs, and the score goes positive. Protecting a lead inverts it. Tied in the first period the two are nearly equal and the score sits near zero.

Tiebreaks. Tied, the leverage term is exactly zero — for a third of all shifts — and thirds of a number that is mostly zero would just be the sign of the score. So two small terms break the tie: who the bench sent (the lineup's offensive-minus-defensive tilt, as a league z-score) and where the puck was dropped. Together they are worth at most eight hundredths against a leverage term that runs from minus one to one, so they settle the label only where the game state does not.

Terciles. The three labels are the thirds of that score over every shift we have, all

seasons together. The two cutoffs are fixed once and stored, so an *attack* shift in 2011 means the same thing as one in 2026, and any one season lands a little off a third each way depending on how many close games it had. An earlier version weighted lineup and zone start at 30 and 15 percent and thresholded at a fixed value that called nine shifts in ten *balanced*; line composition is a consequence of intent, not a cleaner measure of it, and zone start is mostly circumstance, so both are now tiebreaks rather than signals.

What it is not. It is not a measure of what happened — a shift labelled *attack* that gets scored on is still a shift that got scored on. It is context for judging the result, not the result. No outside source publishes coaching intent, so there is nothing to validate this against. Treat it as a lens, and a soft one.

SHIFT VALUE: XG IMPACT AND WP IMPACT

Shift value: xG impact and WP impact

The naming convention. *Ax* means *above expected*. GSAx is the borrowed public name — goals saved above expected — and GAx and SAx follow it: goals and success, each above what the context said to expect. The two shift-value numbers carry plain names instead. **xG impact** is expected-goal differential above what the context predicted; **WP impact** is win probability added above it. They were labelled xGAx and WPAx until October 2026, so an old link or post that says either one means these.

xG impact is the plainer of the two shift-value numbers. Over the selected shifts, take the on-ice shooter-adjusted xG for minus against, and subtract what the league produces in the same contexts — its xGF and xGA per second for the shift's season, strength, zone start, score state, and terciles of competition and teammate quality, times the shift's length. Per 60 it reads as goals per hour

above the league in those spots. It is the same forecast as fWPA below, on the same context cell with the same fallback, valued in goals rather than wins — which is what lets the Shift Performance page's competition filter be read straight: a player facing top-tercile competition is scored against what the league does against top-tercile competition, not against the league's average shift.

On a game's Lines tab the expected side comes from the Linemates context model instead (below, under Linemates): per 5v5 stint, what a league-average five would produce from that zone start, at that score, with that implied purpose, against those five opponents' ratings, with all five of the team's own skaters left out. A unit's xG impact is its xG (sh) for above that plus its xG (sh) against below it, summed over its stints. Leaving the whole five out is what lets every row and column of the tab add up to its total; the price is that a line's figure includes whichever defencemen it played with, which the With-whom view breaks out beside it.

WP impact asks the question a bench actually cares about: what did the shift do to the team's chance of winning, over what a league-average shift in the same spot would have done? Three numbers, all in wins, all summed over whatever shifts you select on the Shift Performance dashboard. They are on-ice numbers, like on-ice xGF: the team's expected change in win probability while the player was out there, not divided among the five skaters who shared it. A shift that changes strength partway (a penalty, its end, a pulled goalie) is valued in pieces, each from its own start at its own strength.

```
WP0  = win probability at the start of the shift, from this bench
xWP  = expected WP at the end, integrating over every chance that occ
fWP  = forecast WP at the end, integrating over the chances the conte
xWPA = xWP - WP0     fWPA = fWP - WP0     WP impact = xWPA - fWPA
```

What WP0 knows and does not. The win probability behind these numbers reads the score, the clock, the strength and the penalty time, and nothing about who is on the ice, so that a shift's value cannot come from the model having already priced the players in it. During a power play it reads how much of the man advantage is left. What it leaves out, by design, is the team: a side leading by two in the first period on the power play wins about five points more often than the state alone says, because that side is usually the better team, and that share belongs to the pre-game number on the Live page, not to the shift.

The chances, not the goals. Each Fenwick shot inside the shift is a coin weighted by its shooter-adjusted xG. The end of the shift depends only on how many of those coins came up for each side, so the expectation is exact: two Poisson-binomial counts and a double sum, no sampling, no first-order shortcut. A shift with no shots still moves WP, because the clock ran in whatever score state it ran in.

The forecast. fWP does the same integration with the league's expected-goal rates for and against in the shift's context — season, strength, zone start, score state, and the terciles of competition and teammate quality — over the shift's actual length. That last choice matters: both halves are valued at the same moment, so the value of time passing cancels in WP impact and nobody is credited for how long the coach left them out.

Reading it. xWPA is leverage-weighted by design: a chance in a tied third period is worth more than the same chance in a 5–1 game, and the win-probability model behind it carries the score, the clock, penalty time still to run and home ice. WP impact is the part the player's shifts added over what league-average shifts would have, and WP impact/60 puts it per hour of ice. Positive means the shifts beat their brief.

Plain WPA. The dashboard also offers WPA/60, the shift valued at what actually

happened: the win probability at the end of the shift from the goals that were scored in it, every goal counted, minus WP0. Nothing is forecast and nothing is taken off, so the clock is in it: a player whose team led while he was out banks the time that ran. WPA minus xWPA is the shift's goals over its expected goals, priced in wins.

Error bars. Shift Performance shows each number with a standard error: the spread of the per-shift residuals divided by the square root of the shift count, rescaled to per 60 by the mean shift length (for SAx, the binomial error on the success share). It treats shifts as independent draws and the league baseline as known, so it measures sample noise, not repeatability.

Why the bars shrink. A raw standard error is only what a player's own shifts can tell you, and 150 minutes of shifts cannot tell you much: the ±2 band on a fourth-liner reached the league leaders. So the number the page shows is a posterior. The league is the prior — the selection's mean, and the spread of its raw values once sampling noise is subtracted out (τ^2 = var(raw) − mean(SE^2), floored at 1% of the raw variance) — and each player is pulled toward it by $B = \tau^2/(\tau^2 + SE^2)$, the share of his own number that survives. Posterior mean $\mu + B(\text{raw} - \mu)$; posterior sd $\sqrt{B} \cdot SE$. The standard normal-normal empirical-Bayes estimator, nothing cleverer. A full-season player keeps four fifths of his raw value; the 150-minute player keeps a third, and his band collapses toward average, because his shifts said little and the league said a lot. The prior is estimated separately for forwards and defencemen, each toward his own group's mean and spread, from every skater in that group who faced the selected QoC/QoT/zone/score/purpose context with at least twenty shifts — the whole context, not just the rows the page happens to be showing. Filtering to forwards, to one team, or raising the minutes floor changes which rows are on screen and never what a player's own number is: those are population filters, not context, and they are applied only after the shrinkage. The tooltip keeps the raw number.

Status. New, and next to SAx on purpose: the two are meant to be compared, and SAx (Success Above Expected, formerly SAE) is under review. Every number inherits the win-probability model's calibration, which is checked by score state, period, strength and home ice before anything here is published.

LINEMATES: A UNIT'S EXPECTED XGD

Linemates: a unit's expected xGD

Together. The Linemates page counts a unit's minutes from *stints*: stretches of the clock with nobody changing on either side and the strength unchanged. Every second of a period belongs to exactly one stint, so a line's time together is an exact sum of whole stints, and a shot counts for the unit that was on the ice when it happened, by the same $start < t \leq end$ rule the shift summaries use, over the same shots: unblocked attempts the xG model scores, with penalty shots left out as play at no strength (so a unit's goals exclude penalty-shot goals). The page keeps 5v5 by count — five skaters and a goalie a side. A line is the three forwards of a unit with exactly three forwards and two defencemen; a pair is its two defencemen; a stint with four forwards out belongs to neither. The *Pick players* view takes any two to five skaters and every stint with all of them on.

The context columns. OZ% is offensive-zone faceoff starts over OZ plus DZ starts (a stint starts in a zone when a faceoff falls within two seconds of its first second; the rest are on the fly, and neutral-zone starts are left out of the ratio). Attack% is the share of the unit's minutes whose stint began in an *attack* state by implied purpose, scored for that bench from the real five on each side. QoC% ranks the TOI-weighted mean shrunk Net Rating of the opposing skaters among the season's 5v5 stint-seconds — 90 means harder competition than nine-tenths of the league's ice time. QoT% ranks the partners outside the unit — the two defencemen behind a line, the three forwards ahead of a pair, the rest of the

five for a pick — among the season's stint-seconds of units of that kind.

Expected xGD/60. Per season, two weighted least-squares fits over the regular season's 5v5 stints, each stint weighted by its seconds:

```
xGF/60 = a + zone + score + purpose + b_own * OWN' + b_opp * OPP'
xGA/60 = a + zone + score + purpose + c_own * OWN' + c_opp * OPP'
OWN' = sum over the team's five skaters of (shrunk Net Rating - leagu
OPP' = the same over the five opposing skaters
```

Zone is OZ, DZ or NZ against on the fly; score is the five score states against tied; purpose is attack or defend against balanced; the league mean is the TOI-weighted mean skater rating of the season, and an unrated skater counts as the mean. A unit's Expected is that fit with the unit's own ratings *left out* of OWN' — what a league-average unit would produce with those partners, against those opponents, from those starts. **xG impact** is the unit's actual shooter-adjusted differential minus Expected, in goals over all its minutes together, and per 60 with one standard error that treats each stint as an independent draw. It is not shrunk toward the league; sort by the total to let minutes count, or raise the floor. A season with under 400 games borrows the previous season's coefficients while using its own ratings and mean. The fitted coefficients are printed under the table.

What is checked. The fit reproduces the season's totals and the zone, score and purpose margins by construction. What can fail, and is checked before a season is shown (`research/linemates_level_check.py`, aggregates in `data/linemates/level_check.json`): the residual by tercile of competition and of teammates and by decile of the own side's rating sum must sit within three standard errors of zero. The page prints each season's verdict from that file, a failure included. A fourth pre-registered rule, that lines with 400 or more

minutes together may not sit more than half a goal per 60 above all lines, was retired by owner ruling 2026-10-13 as mis-specified: Expected is what a league-average unit would do in the same minutes, so the lines coaches keep together are supposed to beat it, and a talent leak into Expected would shrink that gap, not widen it. The gap is still printed. In its place, and reported rather than gated, is the calibration of Expected itself: actual minus Expected summed over every forward line, for the season and by tercile of OZ% and of QoC, both as is and with the lines' own ratings put back. The split-half repeatability of line xG impact/60, and the slope of a line's residual on its own members' ratings (which should be near the fit's own-rating coefficient, since the unit's talent is left out of Expected on purpose), are measured and reported, not gated. Sequence is not cause: a unit that beats Expected did so in those minutes; the number does not say it will again.

IWPA: WHO MOVED THE GAME

iWPA: who moved the game

iWPA — individual win probability added — is an accounting, not a forecast. Every goal, penalty and unblocked shot moves a team's chance of winning by an amount the win-probability model reads off exactly, and so does every stretch of clock in between. iWPA hands each of those moves to the players who were part of it, by a fixed rule, so the credits of one game add up to exactly what happened in it: from the chance at puck drop to the result. Only a shootout stays with the team. Units are wins.

Who gets what. A goal for is split by rungs: scorer 5, first assist 4, second assist 3, every other skater on the ice 1, forward or defenceman alike. A penalty charges the player who took it and credits the player who drew it. The clock between events — protecting a lead, killing a penalty — is shared equally by the

skaters on the ice, once each position's measured share of the game's chances is paid and handed back to the team. A shot's chance is credited to the shooter and debited from the goalie, or the blocker on a block, as soon as it is taken — both at a weight of 5 against 1 for everyone else on the ice. On a miss or a save each side pays its share back equally out of its clock, so the team's total is unchanged while the shooter keeps his larger share. Beyond the chance, a shot is priced in two more stages: reaching the net, which belongs to the defending skaters equally; and on net to goal, which is the goalie's alone and comes to his goals saved above expected, priced in wins. A shot blocked by an opponent is a save by a skater: the blocker is credited the shooter-adjusted expected goals of the shot, averaged over where it most likely came from, times what a goal would have meant, paid out of his own bench's clock. A shot blocked by a teammate counts as a miss.

Reading it. iWPA describes who was on the plays that moved the game and how far each play moved it. It is weighted by leverage: a goal in a tied third period is worth far more than one in a 5–1 game. It does not say who will move the next one, and it is not a talent estimate. For that, read xG impact and Net Rating.

What is not in it yet. Where a blocked shot came from is estimated by a model that fell short of its own pre-registered test; the writeup gives the numbers. Seasons start at 2023–24, the span the win-probability model was trained on; playoffs are out. The number is printed without an interval: it is a sum of observed events priced by one model, so the model's checks are shown in the iWPA writeup instead.

BADGES

Badges

A badge is a claim, not a compliment: that on his last 250 games, the newest weighing most, the player is one of the top ten percent of forwards, defencemen or goalies at the thing named. He earns it by standing in the top 10% of his position group on the posterior score, and only if the probability that he really is a top-decile player at it clears 0.35 — so a badge is never awarded on evidence too thin to support it, and it is kept while that probability holds. Where the evidence is thinnest — a score whose error comes from his own game-to-game spread over fewer than 20 games — the line is 0.50, to earn and to keep, so a hot fortnight is not enough. Because the window runs across seasons, a badge carries into October and a strong start counts at once; a seal shows how long it has been held.

It was a probability line before a rank: a badge went to anyone at better than even odds of being top-decile. The probabilities were right — they sum to about a tenth of every group, in every season — but the share who cleared even odds tracked how precise the badge was rather than how many players deserved it. Minute Muncher, measured at 0.99, named a tenth of the league; Motor, resting on a component that replicates at 0.6, named four in a hundred. Same claim, same calibration, a badge that appeared three times rarer for a reason that had nothing to do with hockey. The rank came first from 2026-09-23, the probability stayed as the floor under it, and the earlier line is recorded here rather than hidden.

There are three kinds, and the difference is drawn on the seal itself, because they are not the same sort of statement:



Skill · an ink disc

A repeatable ability: something he does, that he does again. Every component behind a skill badge replicates at 0.5 or better when a season's games are split in half at random, and that is the only kind gated on it. It is a claim about the player. It is not a claim that the thing helps: Beanbag holders block a lot of shots, reliably, and blocking runs with shots against.

Role · a slate plate

How he has been deployed lately, or how he played it. A fact about his recent games rather than a claim about the man, so it is not gated on replication, and it claims nothing about whether it works. The deployment ones are measured against his own bench, because deployment is a coach dividing one team's ice: Hard Minutes, Minute Muncher, Spearhead, Sentry and Iron Man all ask what share of something his team gave him. Enforcer is the exception, and it is the one that is style rather than deployment — fighting, hitting, sitting in the box and being big are things a player does and is on any roster, so they are scored against the league. Some of them point the wrong way on purpose: Enforcer holders are outscored the following season.

Outcome · a hollow ring

What happened with him on the ice, luck included. Not what he did, and not what he can do. An outcome badge is not expected to repeat — that is what makes it an outcome — so it is not gated on replication either. Scorer counts the goals; Finisher, a skill, is the claim about whether he should have had them. Ledger and Ice Tilter are the same pair in goals and in expected goals, and the gap between them is the luck.

Finisher

F, D · SKILL

Scores more than his shots should: a top-ten-percent shooter on the talent posterior behind GAx(sh); for a defenceman, that posterior together with the shot value he produces himself.

Shooter talent posterior	`weight 1.00 · talent · posterior`
Individual xG(raw) /60, 5v5	`weight 0.78 · own · game`

For forwards, finishing only; says nothing about how many shots he takes. For defencemen the talent posterior alone did not predict next season's goals (r 0.33), and talent plus his own shot value per 60 did (r 0.49, top decile scores twice the rest the next season), so the defenceman's badge is that composite: a shooter who also gets shots.

Forcefield

D · SKILL

Nothing gets through: a top-ten-percent defenceman at xGA below forecast per 60 at 5v5.

xGA below forecast /60	`weight 0.69 · own · shift`

Shot suppression on the ice he was given, not a blocked-shot count. Entry denial is its own badge (Wall).

Wall

D · SKILL

Holds the blue line: a top-ten-percent defenceman at the share of entry attempts against him that he broke up, from All Three Zones tracking.

A3Z entry denial rate, three tracked seasons	`weight 0.68 · All Three Zones · a3z`

Denials over attempts, so it is a rate, not a workload. Defenders who are targeted less deny a little more (r about -0.2 to -0.3 with targets per 60). Pooled over three tracked seasons: one season's ~120 attempts against cannot separate a rate from chance.

Boomerang

D · SKILL

Turns the dumps he goes back for into clean exits: a top-ten-percent defenceman in the share of his retrieval attempts that led straight to an exit, from All Three Zones tracking.

A3Z share of retrieval attempts that led to an exit, league-centred by season, three tracked	`weight 0.65 · All Three Zones · a3z`

seasons	

A rate, not a workload: exits over retrieval attempts (clean and botched), so a defenceman on a team that rarely has to retrieve is not marked down for it. Each season is centred on that season's league rate for defencemen. Not the same as carrying it in (Transporter).



Netfront

F · SKILL

Scores from the blue paint: top-ten-percent in goals per 60 from net-front chances (within 15 feet, tips, deflections and rebounds), counting both how many he gets and how he converts them.

Net-front goals /60: expected goals times his own shrunk finishing, three seasons, all strengths	`weight 0.81 · own · netfront`

Production and conversion in one number. It starts from his expected goals on net-front chances, which replicate (0.81 over three seasons), and multiplies by his own goals-to-expected ratio shrunk toward the league's. The shrinkage is heavy because it has to be: conversion at the net front replicates at 0.17 in a season and 0.25 over three, and the fitted spread of true finishing there is about ten percent. The first version measured production alone and gave Timo Meier gold while he converted at the 20th percentile.



Playmaker

F, D · SKILL

Makes the shot before the shot: top-ten-percent in shot assists and chance assists per 60, from All Three Zones tracking.

A3Z shot assists /60,	`weight 0.78 · All Three Zones · a3z`

three tracked seasons	
A3Z chance assists /60, three tracked seasons	`weight 0.70 · All Three Zones · a3z`

Passing that produced shots. Not points. All Three Zones tracking, shown opaquely: the badge and its probability, never the counts.



Transporter

D, F · SKILL

Carries the puck in: top-ten-percent in controlled entries per 60, from All Three Zones tracking.

A3Z carried entries /60, three tracked seasons	`weight 0.91 · All Three Zones · a3z`

Getting in with possession. Getting out is Boomerang; the two share no component and correlate about 0.2 among defencemen. All Three Zones tracking, shown opaquely.



Motor

F · SKILL

Never stops working: top-ten-percent in forecheck recoveries and penalties drawn, per 60.

A3Z forecheck recoveries /60, three tracked seasons	`weight 0.78 · All Three Zones · a3z`
Penalties drawn /60	`weight 0.63 · own · game`

Work without the puck. Hits and rebounds were tested and rejected: hits run against offence, rebounds do not replicate. Forwards only: the defencemen's score replicated

at 0.31-0.44, under the 0.5 gate.

Speedster

F, D · SKILL

Fast: a top-ten-percent rate of speed bursts over 20 miles an hour per 60 of ice time, from NHL EDGE tracking.

EDGE bursts over 20 mph /60	`weight 0.88 · edge · edge`

Fast, not necessarily good: burst rate is the most repeatable thing measured here (year to year r 0.88) and adds almost nothing to play-driving for forwards (r -0.04) and a little for defencemen (0.22). Penalties drawn do not track it (r 0.13 and 0.04). Cut within position: a median forward bursts three times as often as a median defenceman. Bursts over 22 mph were measured (r 0.83) but are too rare to be a unit (two a season for a median forward), and the 20 mph rate contains them.

Hard Minutes

F, D · ROLE

Sent out against the other side's best, with less help, and on the kill: top-ten-percent in competition faced relative to his own team, in competition minus teammate quality, and in share of his team's penalty-kill time.

Opponent Net Rating faced, relative to his team that game	`weight 0.93 · own · game`
Opponent minus teammate Net Rating, relative to his team that game	`weight 0.95 · own · game`
Share of his team's	`weight 0.98 · own · game`

penalty-kill ice time

A deployment fact. It predicts nothing and is not a compliment. Competition is measured against his own team in the same game, because raw quality of competition carries schedule noise (it replicated at 0.69 for forwards and 0.55 for defencemen; the team-relative version at 0.93 and 0.92). The first version used raw competition alone and missed Nico Hischier, who faces his team's hardest matchups with weaker linemates and kills penalties.



Scorer

F · OUTCOME

Puts the puck in: top-ten-percent in goals per 60 at 5v5 and on the power play.

Individual goals /60, 5v5 + PP `weight 0.60 · own · game`

An outcome, not a talent: it counts the goals he scored, luck and linemates and power-play time included. Finisher is the talent claim -- who SHOULD score, from the shooting posterior behind GAx(sh) -- and the two come apart every season. 5v5 and the power play only: those are the two states a scoring role is played in, and 4v4, 3v3, the kill and empty-net play are deployment, not scoring. It repeats poorly -- the rate agrees with itself at 0.60 across random halves of a season -- and it is published anyway, because an outcome badge is not a claim that the player will do it again. Forwards only, and no longer for the reliability: defencemen were tried in 2026-09-23 under the outcome rule and the level check threw them out. A defenceman's goal rate is so noisy that almost nobody clears even the 0.35 confidence floor, so the badge named 1.3% of eligible defencemen in 2023-24 and 3.2% in 2024-25 against a floor of 4%. A badge that goes to three players in a league of six hundred is not a top-decile claim, whatever it is measuring.



Ledger

F, D · OUTCOME

What the scoreboard did with him out there: top-ten-percent in on-ice goals for minus

against per 60 at 5v5, against what his team managed with him on the bench.

On-ice GF - GA /60 at 5v5, relative to his team off his ice	`weight 0.42 · own · game`

This is what happened with him on the ice, not what he did: his goalie's night, his shooters' night and the bounces are all in it. The relative term takes out the team he plays for, not the line he plays on. It barely repeats -- split a season's games in half at random and the two halves agree at 0.47 for forwards and 0.36 for defencemen -- and that is the badge working, not failing: a goal ledger is mostly things that happened to a player. Ice Tilter asks the same question in expected goals, which repeat at 0.85, and the gap between the two numbers is the size of the luck. Read them together.



Minute Muncher

F, D · ROLE

Plays the most: top-ten-percent in the share of his own team's ice time at his position.

Share of his team's all-strength ice at his position group, per game	`weight 0.99 · own · toi_share`

A deployment fact, and the most repeatable number on the site (0.99 split-half): it says the coach trusts him, not that the coach is right. It is his share of the ice his own bench got at his position, not raw minutes (CJ, 2026-09-23: the role badges should be relative to team). That turns out to change almost nothing -- the share and the raw rate correlate 1.00 within a position group across all three seasons -- because a team's forwards share about 180 minutes a night and its defencemen about 120 whatever kind of team it is, so the denominator is nearly a constant. It is the right denominator anyway, and now the badge says what it always meant: his coach leans on him. Ice time does track play-driving -- r 0.62 with xG impact among forwards, 0.30 among defencemen -- but that correlation runs through the coach's judgement, not through anything this badge measures, so it is his opinion with a seal on it and not an independent one. Cut within position, because a defenceman's share of a two-pair-

deep group is twice a forward's share of a four-line one.

Spearhead

F, D · ROLE

Sent out to attack: top-ten-percent in how much more than his ordinary share of his team's ice he is given when the situation calls for attacking.

Share of his team's 5v5 attack-purpose ice over his share of the rest, per game, pooled	`weight 0.68 · own · share`

Deployment, not outcome, and relative to his own bench. It compares, game by game, his share of his team's attack-purpose 5v5 ice with his share of the rest of its 5v5 ice, and pools the games (a Mantel-Haenszel ratio): one when the bench turns to him for it exactly as often as it turns to him for everything else, above one when it reaches for him to chase a goal. A game in which his team never needed to attack adds nothing to either side. Implied purpose reads the win-probability asymmetry of the shift's starting state, with the lineup and zone as a tiebreak; it is where the coach put him and what the scoreboard asked for, and it says nothing about whether he attacked well. It used to be the raw share of his OWN shifts that carried an attack purpose, and that was a fact about his team before it was a fact about him: a team that trails a lot attacks a lot, and the badge went 60% to a single roster in 2025-26 while correlating -0.47 to -0.69 with team win share (CJ, 2026-09-23). The two versions barely agree -- 0.14 to 0.26 across the seasons and groups -- so this is a different badge, not a rescaled one. It also replicates BETTER than the version it replaces, 0.68 for forwards and 0.68 for defencemen against 0.46 and 0.43, which is not what was expected: taking the team out removes a large stable term, and the badge repeats more anyway, because what is left is a real and persistent difference between the men on one bench. It is not gated on that number either way -- a role badge records how a season was actually played, and that is true whether or not the next half-season repeats it.

Sentry

F, D · ROLE

Sent out to defend: top-ten-percent in how much more than his ordinary share of his team's ice he is given when the situation calls for defending.

Share of his team's 5v5 defend-purpose ice over his share of the rest, per game, pooled	`weight 0.51 · own · share`

Deployment, not outcome, and the mirror of Spearhead: his share of his team's defend-purpose 5v5 ice against his share of the rest of its 5v5 ice, game by game, so a game his team never defended counts for nothing. It says the bench turned to him to hold a lead or kill a shift more often than it turns to him in general, not that he held it. Hard Minutes is the badge for who he was sent out against. The raw version -- the share of his own shifts that carried a defend purpose -- was mostly a measurement of his team: winning teams protect leads, so the badge piled up on the good ones, and its per-team earner count correlated +0.37 to +0.55 with team win share (CJ, 2026-09-23). Two random halves of a season agree at 0.59 for forwards and 0.42 for defencemen, against the raw share's 0.48 and 0.41: better for forwards, for the reason Spearhead gives, and level for defencemen. The pooled-share form it replaced on 2026-10-09 repeated more (0.65 and 0.56), and part of that was the flaw repeating: a player who plays little in games his team never defends does so every half-season. A role badge is not gated on that either way, because how a season was played is a fact about the season.



Enforcer

F, D · ROLE

The heavy: top-ten-percent in a mix of fighting majors, hits, penalty minutes and size, with the fights weighted heaviest.

Fighting majors, sqrt, within position	`weight 1.00 · own · season`
Hits /60, all strengths	`weight 1.00 · own · game`
Penalty minutes /60, all strengths	`weight 0.73 · own · game`

Height and weight, standardised within position	`weight 1.00 · own · constant`

A style fact, and not a flattering one -- style, not deployment, which is why it alone among the role badges is not measured against his own team (CJ, 2026-09-23). Sentry, Spearhead, Minute Muncher, Hard Minutes and Iron Man all ask what share of something his coach gave him, and a share needs a bench to be a share of. Fighting, hitting, sitting in the box and being big are things a player does and is, on any roster, so they are scored against the league's position group and not against his teammates. The fight rate runs slightly NEGATIVE against play-driving, about -0.2 with xG impact, and the badge answers for that every season: forwards who hold it go on to a 44-46% on-ice goal share the following year against 50% for everyone else (p 0.001 and 0.011 in the two seasons that can be tested). Nothing here claims fighting helps, and the numbers agree. You do not have to fight to earn it -- every skater is scored, and a big man who hits hard and lives in the box can get there without dropping the gloves once -- but the fight is what the badge is named for, so it is weighted level with hitting and with size, 27% of the score each, and penalty minutes carry the rest at their measured 20%. Those three equal weights are chosen rather than measured -- fighting replicates at 0.72 and hits at 0.97 -- and they are the only chosen numbers in the badge system. The fight count enters as its square root, so the first fight is worth most and a tenth one cannot run away with the badge.



Beanbag

F, D · SKILL

Stands in the lane: top-ten-percent in the share of the shot attempts against him that he blocks himself, at 5v5 and on the penalty kill.

Blocks made per on-ice shot attempt against, 5v5 + PK	`weight 0.75 · own · share`

A skill badge means a repeatable ability, not a useful one: blocking at this rate replicates at 0.77, so it is something the player reliably does, and it still runs WITH shots against, so it is not something that helps his team. Both are true at once and the

seal only claims the first. A rate, not a count, and that is the whole change from the first version: blocks per 60 rewarded being hemmed in, because a player in his own end gets more chances to block. Per attempt against, the denominator is the siege itself, so what is left is how often he is the one who steps in front. Still not a skill claim and still not connected to outcomes: the share is a style and a position, and Forcefield is the badge for actually suppressing the chance. 5v5 and the kill only -- the power play carries two percent of the attempts against, and a block there is an accident.



Iron Man

F, D · ROLE

Never out of the lineup: a top-ten-percent share of his team's games played over the last three seasons.

Games played / team games, three seasons	`weight 1.00 · own · binomial`

Availability, which is a fact and not a skill, and the only badge here that deliberately looks past the current season. A player is scored only if he appears in all three seasons of the window: a rookie who played all 82 has not yet made the claim, so he is not scored rather than badged at a perfect share. Traded seasons count his games against his primary team's schedule, not two teams' schedules. A role badge and not an outcome: being available does not predict goals, and this one says nothing about whether he should have been in the lineup -- only that he was in it.



Quick Draw

F · SKILL

Wins the draw that matters: a top-ten-percent centre at faceoffs, adjusted for who he was up against and for what each draw was worth.

Stake-weighted Bradley-Terry faceoff rating	`weight 0.86 · own · draw`

It is faceoff percentage, adjusted -- and the adjustments move very few centres: the rating correlates 0.93 with raw faceoff percentage, and it repeats about as well (0.86 across random halves of a season against 0.86 for the raw number). That is worth saying plainly rather than dressing up. The adjustments are real but small because every centre takes a similar mix of draws: an end-zone draw at even strength swings 0.0157 expected goals over the next fifteen seconds and a neutral-zone one 0.0059, a ratio of under three, so weighting by what a draw is worth reorders almost nobody. Winning draws is also not the same as being worth ice: this says nothing about the rest of his game.

Ice Tilter

F, D · OUTCOME

Tilts the ice: xG impact/60 at 5v5, both halves, against what the league does in the same spots.

xGF above forecast /60	`weight 0.90 · own · shift`
xGA below forecast /60	`weight 0.66 · own · shift`

Outcome, in progress.

An outcome, not a skill: the mechanism badges explain about two thirds of it. Outcome badges are still being worked out.

The full account, with the validation tables and the correlations between badges, is in the badges writeup.

One state, no tiers. Earned or not, never a letter grade, never a percentile bar. The top three in each badge and position group wear gold, silver and bronze, which is a rank among holders, not a grade. A percentile compresses the tails and invites reading a rank into a number that doesn't support one; a badge is a yes-or-no claim with the evidence attached, and the shrinkage does the ranking's job honestly.

Thresholds. The top 10% of the group, and 0.35 of probability to earn and to keep (0.50 where that rests on his own game-to-game spread over fewer than 20 games) are the only chosen numbers in the whole system. Everything upstream of them — the shrinkage, the posterior, the sample floor — is fit, not picked. The floor is where a player's own evidence starts to outweigh the position-group prior: below it, the badge is mostly describing the league, not the player, so it stays off rather than claim more than the shifts support.

HOURLY BURST

Hourly burst

NHL EDGE publishes three counts per skater-season: bursts that peaked at 18 to 20 mph, 20 to 22, and over 22, plus the ten fastest speeds of the season. Those are the top three bars of a histogram we never see the rest of. The player page turns them into one curve: how many times per 60 minutes of ice time the skater exceeds a given speed. Height at the left is how hard he skates (sprints over 18 per 60); reach to the right is how fast he is.

In one sentence: we fit a distribution to each player's bursts (his EDGE counts above 18, 20 and 22 mph plus his ten fastest speeds), with one shared shape per position and a fitted speed level for each player, shrunk toward his position's average. The headline is his implied hourly burst: the top speed he reaches about once per 60 minutes of ice time.

Hourly burst	The speed at which the fitted curve crosses once per 60 minutes of ice time: the pace he reaches about once per hour on the ice. EDGE's own max speed is the pace he reached once all season, a single reading; this is the same idea read from the whole season, and it is far

	steadier year to year.
Sprints over 18 per 60	The three bins added up, per 60 of all-situation ice time. A role and effort figure as much as a skating one: forwards who sprint most tend to be energy-line forwards with fewer minutes and less power play.
The fit	Each skater's burst peak speeds are treated as draws from a reversed Gumbel distribution, a bell with a light upper tail, which is what a physical top speed looks like. One width per position, fitted on 2021-22 to 2025-26; the skater sets only where the distribution sits, read from the ratio of his bins and from his ten fastest speeds, shrunk toward his position's mean. The alternatives were tried and rejected on one test: predicting the season's max and tenth-fastest speed from the bins alone, by decile. A Gaussian tail put the max a third of a mile an hour too high for every skater. A Generalized Pareto over 18 mph, the textbook model for speeds over a threshold, and a Weibull were later run through the same test season by season; neither beat it.
Typical burst	The fit's location: his most common burst peak, in mph. It is inferred, because EDGE sees only the bursts over 18 mph, so for most skaters it sits below 18 and is read off the shape of what shows above. The hover gives an 80% range.
Burst spread	The fit's scale: how spread out a skater's burst peaks are around his typical burst. It is not his own. Every forward shares one, 2.8 mph for forwards, and every defenceman another, 3.1 mph for defencemen, fitted on the

	league; the data do not support a width per skater. The standard deviation of his burst peaks is about 1.28 times the scale: 3.6 mph for a forward, 4.0 for a defenceman.
What is drawn	His curve with an 80% band, his three bin rates as circles at 18, 20 and 22 mph, and his ten fastest speeds as small dots placed at the rate they imply: the k-th fastest speed of a season is, by definition, a speed exceeded k times that season, so it sits at k per hours of ice. The grey bands hold the middle 20, 50, 80 and 96 percent of skaters at his position that season.

Fast is not the same as good, and the two dimensions mean different things by position. For forwards the top end predicts controlled zone entries, rush shots and individual expected goals, while sprint volume predicts deployment. For defencemen it is the reverse: the ones who sprint often are the ones who carry the puck. Neither feeds Net Rating. Shown for skaters with at least two hours of ice in the season; the current season updates nightly.

ALL THREE ZONES MICROSTATS

All Three Zones microstats

Hand-tracked by Corey Sznajder for a sample of each season's games (about a third, 2021-22 onward). Licensed to us; shown only to active patrons of his Patreon, on the All Three Zones page. Every count is 5v5 and per player.

Two families, and they answer different questions. The *expected box score* predicts what a player's real NHL line should look like, and is calibrated against it. Everything below it counts what Corey's tracking recorded. The two are not

interchangeable: a pass he did not record cannot be credited to anyone, so the tracking-derived *xG created* columns sit about a third below real assist totals by construction. That is a property of the data, not a flaw in it, and it is why these are separate sections rather than one.

xG	Expected goals from the player's own 5v5 shots, calibrated against goals actually scored.
xA1	Expected primary assists, calibrated against real primary assists.
xA2	Expected secondary assists. The noisiest of the family (season r 0.81) -- a secondary assist depends heavily on who else was on the ice.
xA	Expected assists, primary and secondary together.
xP	Expected points: goals plus both assists. The 80% band contains the player's real 5v5 point total about four times in five.
xP1	Expected primary points: goals plus primary assists, the version that survives secondary-assist noise.

The expected box score is a full-season projection, so it does not change with the tracked/estimated toggle. Each cell shows an 80% interval containing the player's real total about four times in five; the point estimate is on hover and is what sorts. It is left blank below five tracked games, where it runs about 15% high — an empty cell rather than a number we know to be wrong.

Entries	Offensive-zone entries by the player, any type (carry, pass, dump).

Carries	Entries with possession: carried or passed across the blue line.
Carries w/ chance	Controlled entries that produced a scoring chance.
Dump-ins w/ chance	Dump-in entries that produced a scoring chance.
Failed entries	Entry attempts broken up at the line.
Targets	Entry attempts against the player as the defender.
Denials	Entry attempts the player broke up at the line.
DZ retrievals	Defensive-zone puck retrievals.
Retrievals → exit	Retrievals that led directly to a zone exit.
Botched retrievals	Retrievals the player lost.
Recoveries	Opponent dump-ins the player recovered on the forecheck.
Exits	Defensive-zone exits by the player, any type.
Exits w/ possession	Exits that kept possession: carried or passed out.
Failed exits	Exit attempts that turned the puck over in the zone.
Shots	Shot attempts, as Corey records them.
Shot assists	Primary passes that led to a shot attempt.
Chance assists	Primary passes that led to a scoring chance.
xG created (1st pass)	The expected goals on the unblocked shots the

xG created (1st pass)	The expected goals on the unblocked shots the player set up with the primary pass, counted where Corey recorded him as the passer. Joined through the shot, not the pass -- the play-by-play records no passes at all. This is NOT expected assists: it runs about a third below real 5v5 primary assists, because a pass he did not record cannot be credited.
xG created (2nd pass)	The same idea one pass further back. Corey records a second passer on 37.7% of tracked 5v5 shots against 74.1% for the primary, so this is a thinner sample, a wider band, and further below real secondary assists.
xG created (3rd pass)	Expected goals on shots where the player made the third-last pass. No point is awarded for this in hockey. Recorded on 12.6% of tracked 5v5 shots and much the weakest of the three: on held-out tracked games a forward's season total correlates 0.56 with the truth, against 0.87 for the first pass, and his two halves of a split season agree at only 0.29. Read it as a wide-banded style indicator, not a ranking.
Rush shots	Shot attempts off the rush, as Corey tags them.
Rush shot assists	Primary passes that led to a shot off the rush.
Forecheck & cycle shots	Shot attempts off the forecheck or cycle, as Corey tags them.
Forecheck & cycle shot assists	Primary passes that led to a shot off the forecheck or cycle.
Carries allowed	Entries against the player that were carried in with possession.
Passes allowed	Entries against the player that were passed

Passes allowed	Entries against the player that were passed in.
Carry chances allowed	Carry-ins against the player that produced a scoring chance.
Dump chances allowed	Dump-ins against the player that produced a scoring chance.
High-danger passes	Primary passes into a high-danger area that produced a shot.
Home-plate passes	Primary passes into the home-plate area.
Centre-lane passes	Primary passes through the middle lane of the ice.
Low-to-high passes	Primary passes from below the dots back to the points.
Behind-net passes	Primary passes from behind the goal line.
Neutral-zone passes	Primary passes made from the neutral zone.
Defensive-zone passes	Primary passes made from the player's own zone.

Carry %	Share of the player's entries made with possession (carried or passed in).
Entry → chance %	Share of entries, controlled or dumped, that produced a scoring chance.
Failed entry %	Share of entry attempts broken up at the line.
Denial %	Share of entry attempts against the player that he denied.
Carry-against %	Share of entries against the player that came in with possession.

Chance-against %	Share of entries against the player that produced a scoring chance.
Botched %	Share of retrievals the player lost.
Retrieval → exit %	Share of retrievals that led straight to an exit.
Possession exit %	Share of exits that kept possession.
Failed exit %	Share of exit attempts that turned the puck over.
Chance share %	Share of the player's shot assists that set up a scoring chance.
High-danger pass %	Share of the player's shot assists that were high-danger passes.
xG created per shot assist	Average danger of the chances the player creates: expected primary assists over shot assists. Note the denominator counts every attempt he set up while the numerator counts only unblocked ones, so this reads lower than the danger of a pass that gets through.
Rush share %	Of the player's shots off the rush or off the forecheck and cycle, the share that came off the rush. A style number, not a quality one.
PP carry %	Share of the player's power-play entries made with possession.
PP setup %	Share of the player's power-play entries that set up a shot attempt.

Special teams are Corey's power-play and penalty-kill counts, added at his request. They are tracked sums only, the same number under both views: no

estimate exists for the games he did not track, and a per-60 there divides by the player's or team's tracked power-play or penalty-kill time, never 5v5 time. On the team tables a team's power-play columns *for* are over its own power-play time and *against* over its penalty-kill time; the penalty-kill columns the other way round.

PP entries	Offensive-zone entries on the power play, any type.
PP carries	Power-play entries with possession: carried or passed across the blue line.
PP entry setups	Power-play entries that set up a shot attempt (Corey's '5v4 Setups'; his sheet rates them as a share of entries).
PP shots	Shot attempts on the power play, as Corey records them.
PP passes	Passes completed on the power play, any kind.
PP shot assists	Primary passes that led to a power-play shot attempt (Corey's '5v4 Primary Setups').
PP chances	Power-play scoring chances, as Corey tags them.
PP chance assists	Primary passes that led to a power-play scoring chance.
PP cross-slot passes	Power-play passes across the slot that led to a shot.
PP low-to-high passes	Power-play passes from below the dots back to the points that led to a shot.
PP goal-line passes	Power-play passes from the goal line that led to a shot.

PK entries	Offensive-zone entries while killing a penalty.
PK carry denials	Power-play carry-ins the penalty killer broke up at the line.

Tracked games are Corey's raw counts over the games he tracked; per-60 and per-84 there are the player's pace over those games only, and the *Tracked* and *Cov %* columns say how many that was. **Estimated season** adds a model's expectation for the untracked games and reports an 80% interval; sorting uses the point estimate. Method and validation are in the tracking-data writeup.

SHIFT SUMMARIES (THE FOUNDATION)

Shift summaries (the foundation)

One row per shift segment: a player's shift, cut wherever his team's strength changes, so a shift that runs from five-on-five into a power play is two rows. Strength is read second by second from who is on the ice (from the play-by-play's situation code where a shift chart is incomplete), the same reading our time-on-ice by strength uses, so a pulled goalie counts the moment he leaves. Seasons before 2026-27 still hold the old rows, one strength per shift taken at its start, until they are rebuilt. The substrate the on-ice metrics are built on, and the foundation for implied purpose and win-probability-added. The schema:

```
shift_id, game_id, player_id, team_id, period,
start_seconds, end_seconds, toi_seconds,
strength_state, zone_start, score_diff_at_start,
teammates_on_ice[], opponents_on_ice[],
cf_for, ca, ff_for, fa, gf, ga,
xgf_fully, xga_fully, xgf_*adj, xga_*adj,
```

```
xgf_shooter_adj, xga_unadj,
implied_purpose?, net_rating_delta?  // future
```

`zone_start` resolves by matching the segment's start (±2s) to the nearest faceoff event; if no matching faceoff, `zone_start = "fly"`. Offensive or defensive depends on which end the team is attacking that period, which is read game by game from the play-by-play (its own zone codes, then shot locations) rather than assumed from the period number. *On-the-fly* changes are the majority (~57%); the rest split roughly evenly across NZ/OZ/DZ.

See `backend/docs/shift-pipeline.md` for the remaining planned extensions (net_rating_delta).

COLOPHON

Built on FastAPI + PostgreSQL, with the shot model fit by a hand-rolled penalised regression rather than an off-the-shelf one, and Next.js + Recharts on the front. Type set in Lora and Inter, with JetBrains Mono for tabular figures. Source available; this glossary renders live from the database, so the numbers above are the same ones the leaderboards use.

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